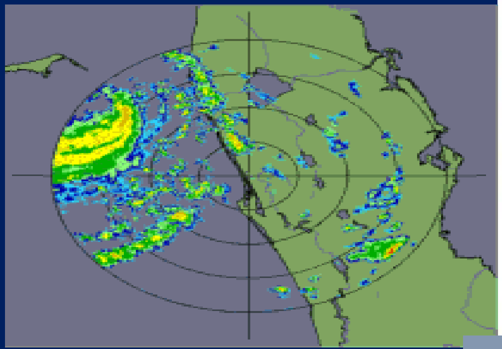
USPCAS-W
Mehran University

HYDROINFORMATICS

GIS Data Sources and Acquisition

By: Dr. Arjumand Zaidi

Senior Research Fellow



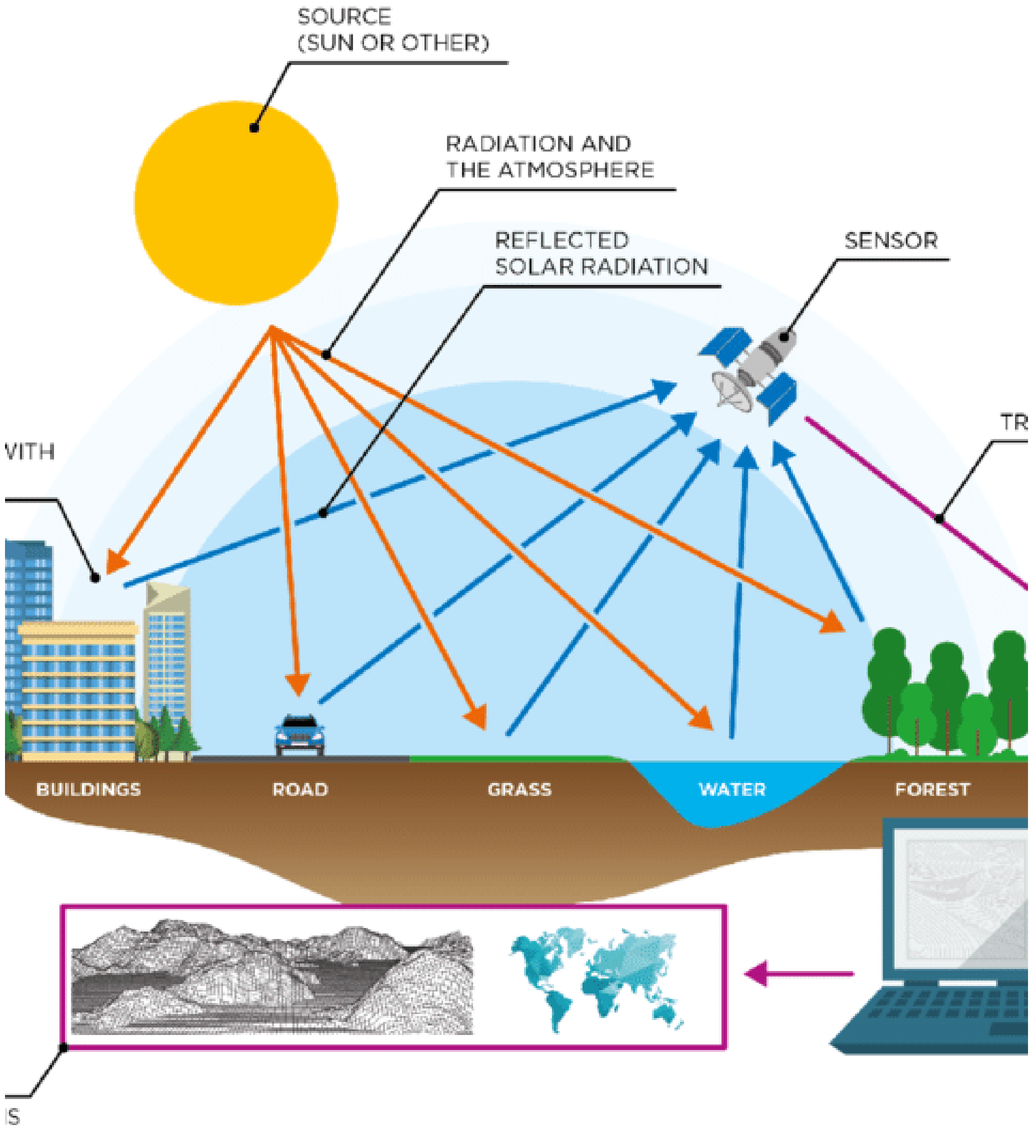


Fall 2026

US Pakistan Center for Advanced Studies in Water

MUET, Jamshoro

Let's kickstart our brains on this topic.



- What are some common on-the-ground methods for data collection?
- What are the primary sources of satellite data for Earth observation?

Learning Objectives

1. To discuss data acquisition challenges.
2. To learn about free satellite data sources.

Data Collection Methods

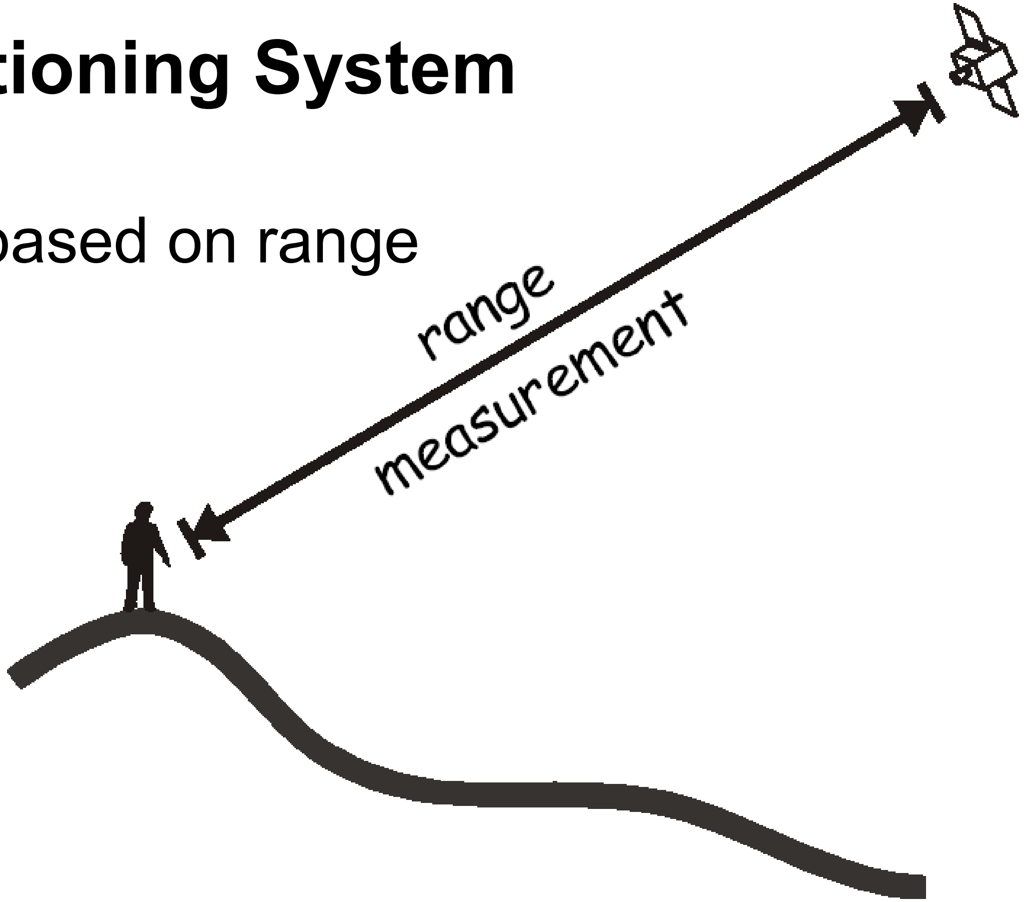
- **Secondary Data Collection**
  - existing data
  - already published/available
  - Someone else collected data
- **Primary Data Collection**
  - from scratch
  - You collected or assigned someone to collect

Field Measurements

Coordinate Surveying

GPS: Global Positioning System

Position is estimated based on range measurements.



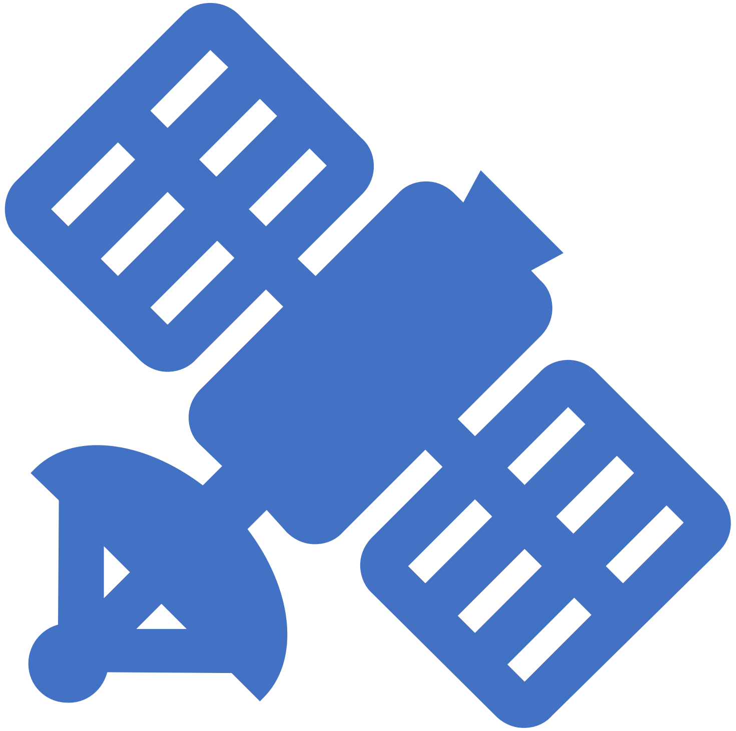
Range = speed of light x travel time

Range = c(t/2)

(c =299,792,458 meters per second, t = round trip time of a signal)

Remote Sensing: measuring electromagnetic energy reflected or emitted from objects – airborne or satellite-based instruments

satellites vary by: orbit, altitude, resolution, revisit frequency, revisit variability (steering) capability, width of swath, image size, stereo capability, wavelengths collected, other sensors, customer delivery time, etc.



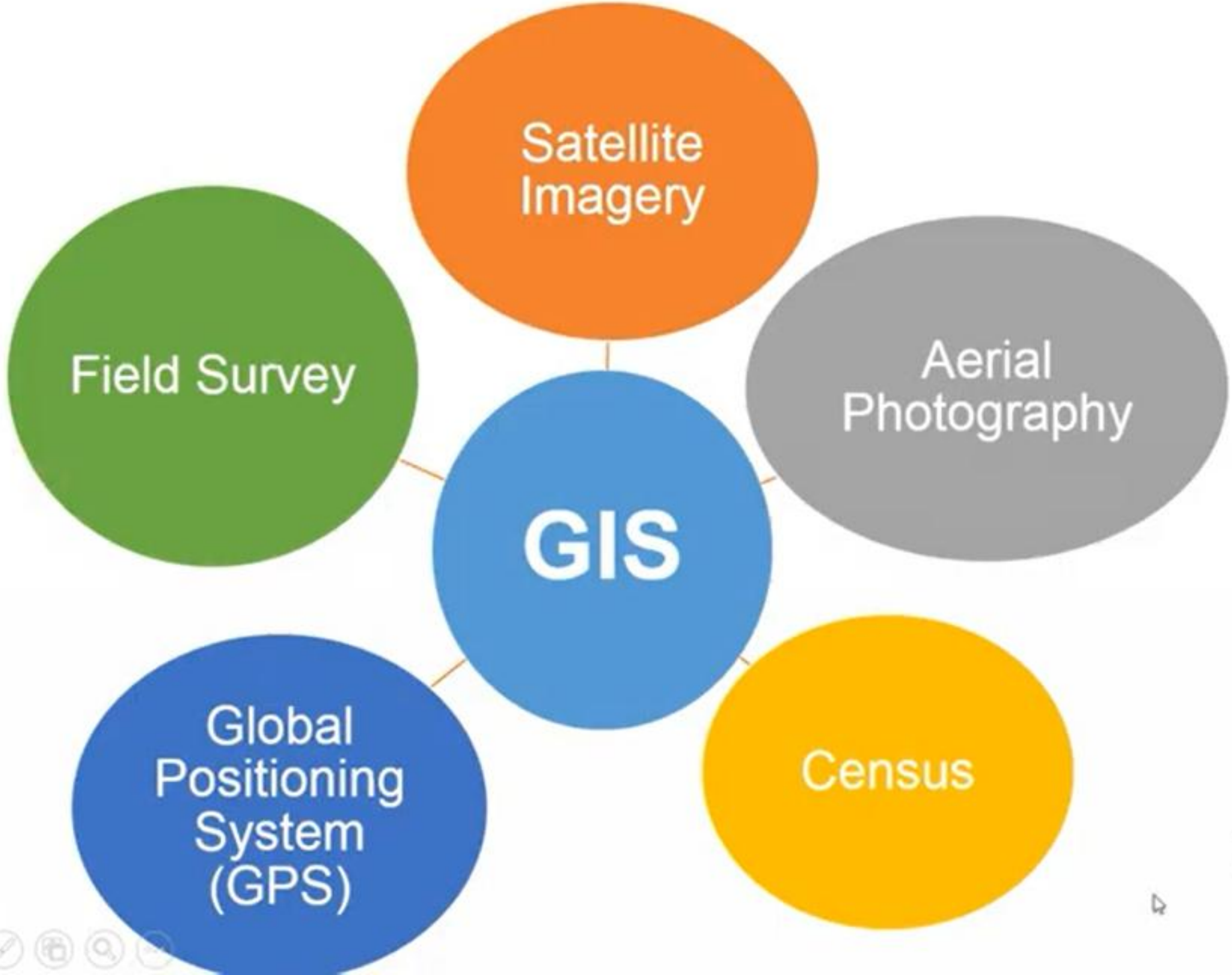
Data Collection Protocols

- Employ a reliable sampling method for representative data.
- Standardize procedures and employ quality control to enhance data reliability and consistency.
- Minimize bias in primary data collection.
- Obtain consent and safeguard participants' confidentiality and rights.

Data use protocols

- Credit secondary data sources through proper citation.
- Ensure data is obtained and used ethically.
  - Adhere to pertinent laws/regulations to ensure legal conformity, safeguard intellectual property rights, and uphold ethical standards.
- Assess the quality and reliability of data before utilization.

Sources of GIS Data



GIS Data Sources

Maps and Drawings

- Scanning or digitizing

Aerial/Satellite Images

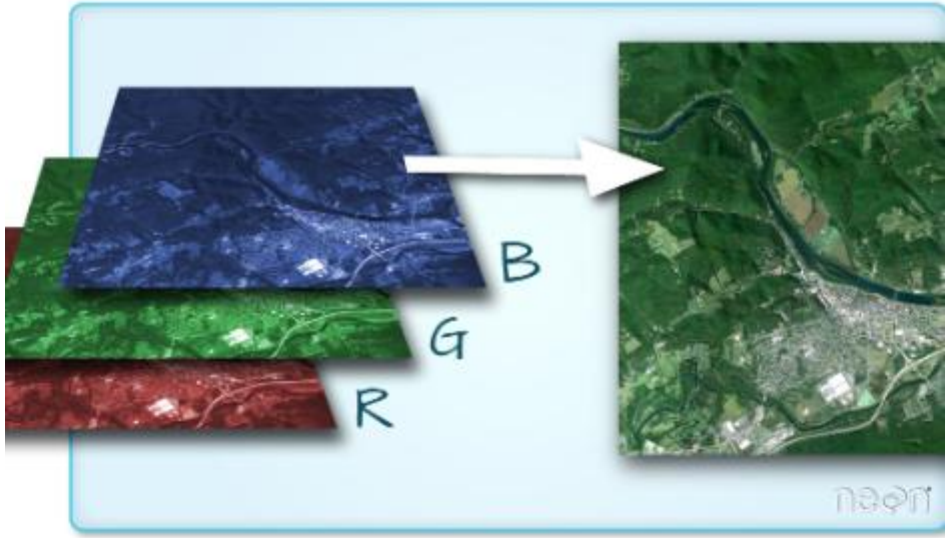
- Image interpretation for features extraction
- Digital terrain model) to create digital orthos

CAD Databases

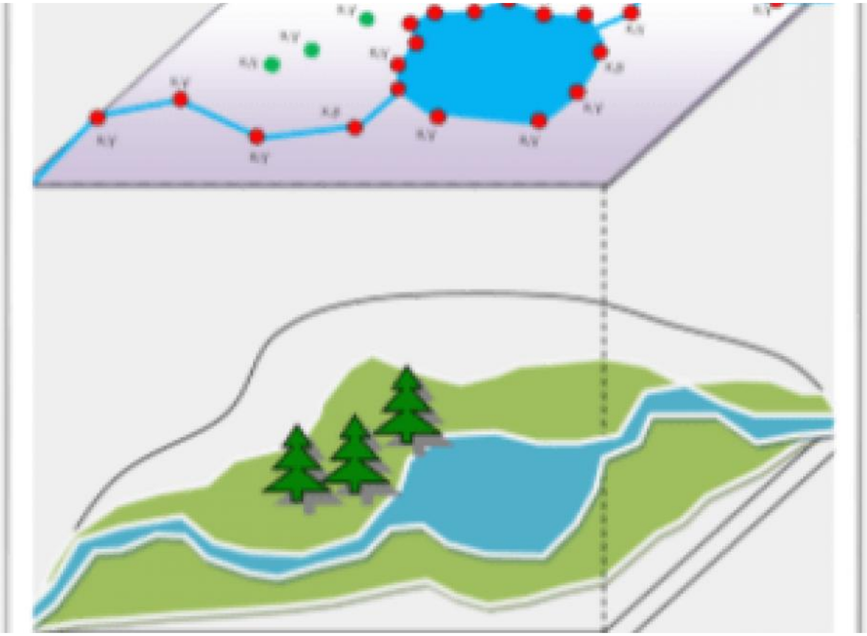
GIS Databases

Attribute Databases

Records and Documents



Sources of Vector/Raster Data



esri Products Industries Support & Services Stories About

ArcGIS Open Data

Steer decisions and results with your data

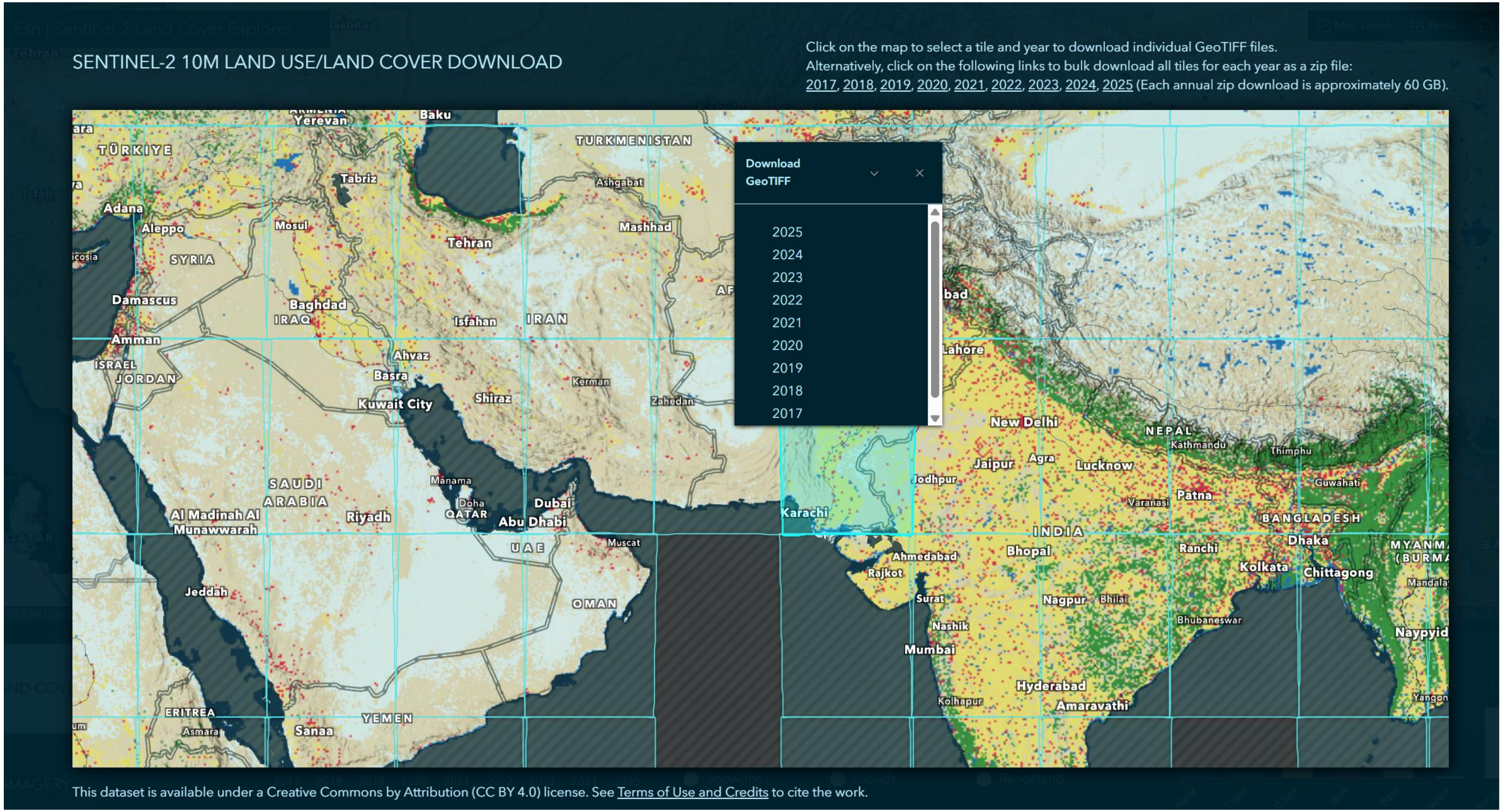
Data is critical to decision making. Whether you are buying a house, wondering about road construction, or opening a new business, trustworthy information is essential. Use Esri's SaaS open data capabilities to provide your community with the authoritative data they need to make better decisions and accomplish their goals.

https://www.esri.com/en-us/arcgis/products/arcgis-open-data

Landcover Explorer

https://livingatlas.arcgis.com/landcoverexplorer/

ESRI LULC 2017 - 2025



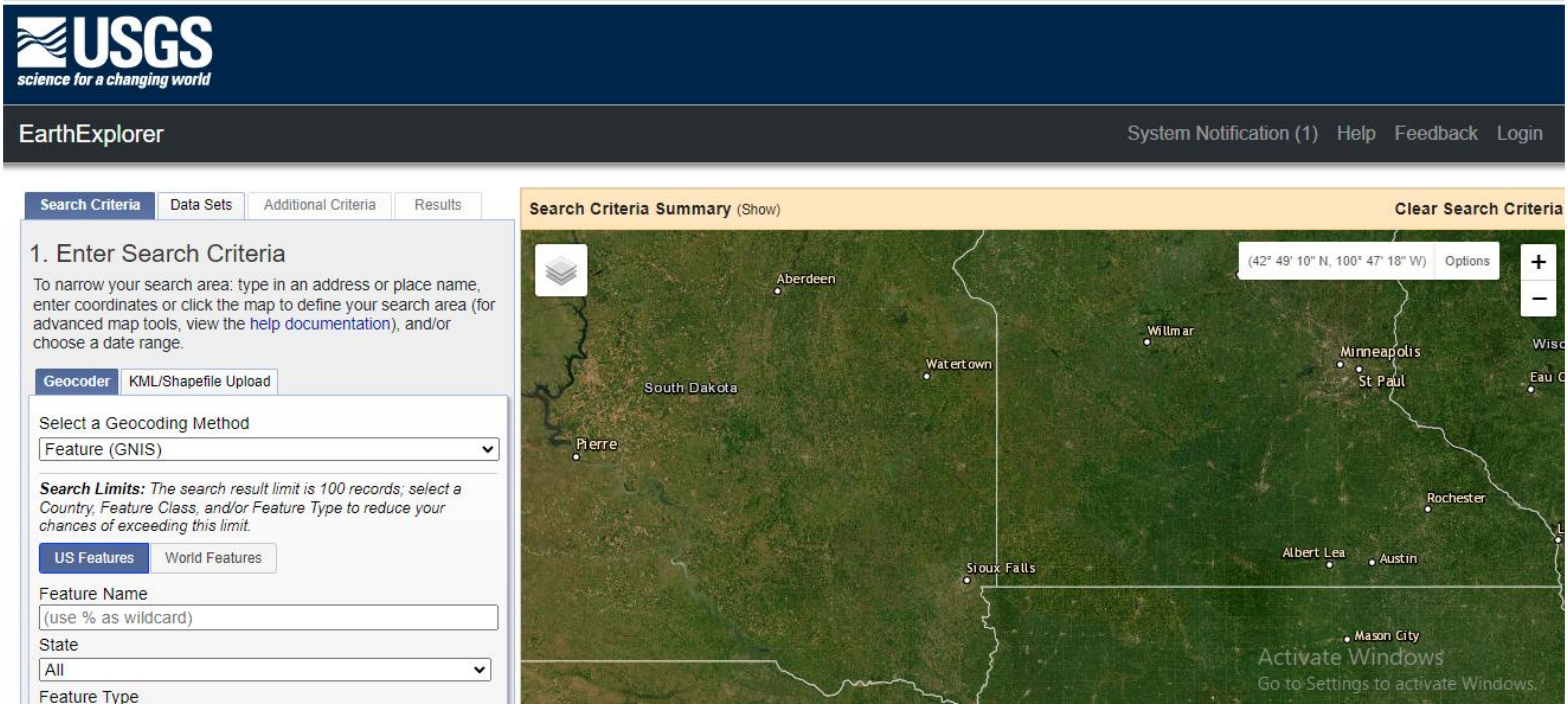
https://livingatlas.arcgis.com/landcoverexplorer/#mapCenter=70.32492%2C30.11343%2C5.88&mode=step&timeExtent=2017%2C2025&year=2025&downloadMode=true

Natural Earth Data

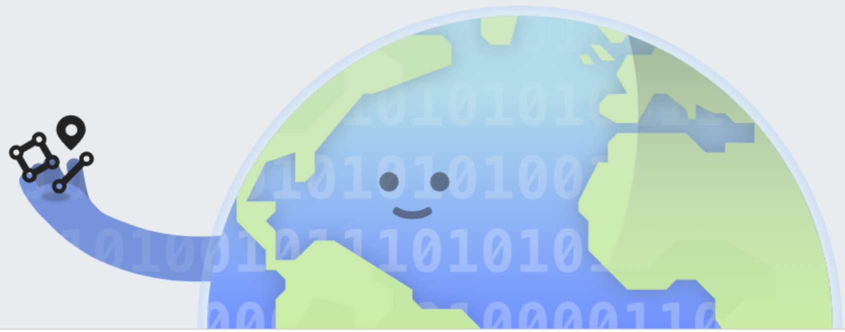
Link: https://www.naturalearthdata.com/

USGS Earth Explorer

Link: https://earthexplorer.usgs.gov/



https://www.openstreetmap.org/#map=6/30.67/69.36



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Free and editable. Unlike other maps, OpenStreetMap is completely created by people like you, and it's free for anyone to fix, update, download and use.

Sign up to get started contributing.

Email

Your address is not displayed publicly, see our privacy policy for more information.

Display Name

Your publicly displayed username. You can change this later in the preferences.

Password

Confirm Password

Verify you're a human

Success! CLOUDFLARE Privacy • Help

By signing up, you agree to our Terms of Use, privacy policy and contributor terms.

Sign Up

Another Option for Downloading OSM is from GeoFabrik Downloads https://www.geofabrik.de/data/download.html

Our Download Server

Our download server at https://download.geofabrik.de has excerpts and derived data from the OpenStreetMap dataset available for free download. Most of these files are updated every day – any change you upload to OpenStreetMap should be on our download server the next day.

Geographic Regions

Data on the download server is organised by region. The root directory contains files that have a whole continent's data in them, and for all continents except Antartica there are subdirectories in which you find individual files for various countries. Some countries again have their own subdirectories with data for administrative subdivisions. For Germany, all "Bundesländer" are available as separate files, and for England we have all individual counties. For France, we have all (historic) Departements. The file for Germany's Baden-Württemberg region is, for example, at

```
https://download.geofabrik.de/europe/germany/baden-wuerttemberg-latest.osm.pbf
```

Please let us know if you are interested in other regional excerpts that we do not offer right now – we might be able to add them with little effort.

Raw data (OSM source data)

Files with the `.osm.pbf` suffix are OpenStreetMap raw data files in the PBF format. They are the common format for OpenStreetMap raw data and need few memory for storage. Daily diff updates (`.osc.gz` , Gzip compressed OSM XML) are available on our download server as well. They contain the changes of one day. The PBF files can be updated with them.

The Bzip2 compressed XML files we used to have on the download server are now discontinued since almost all software supports .osm.pbf. If you still need bz2 files, you can create them yourself, or order them commercially from us.

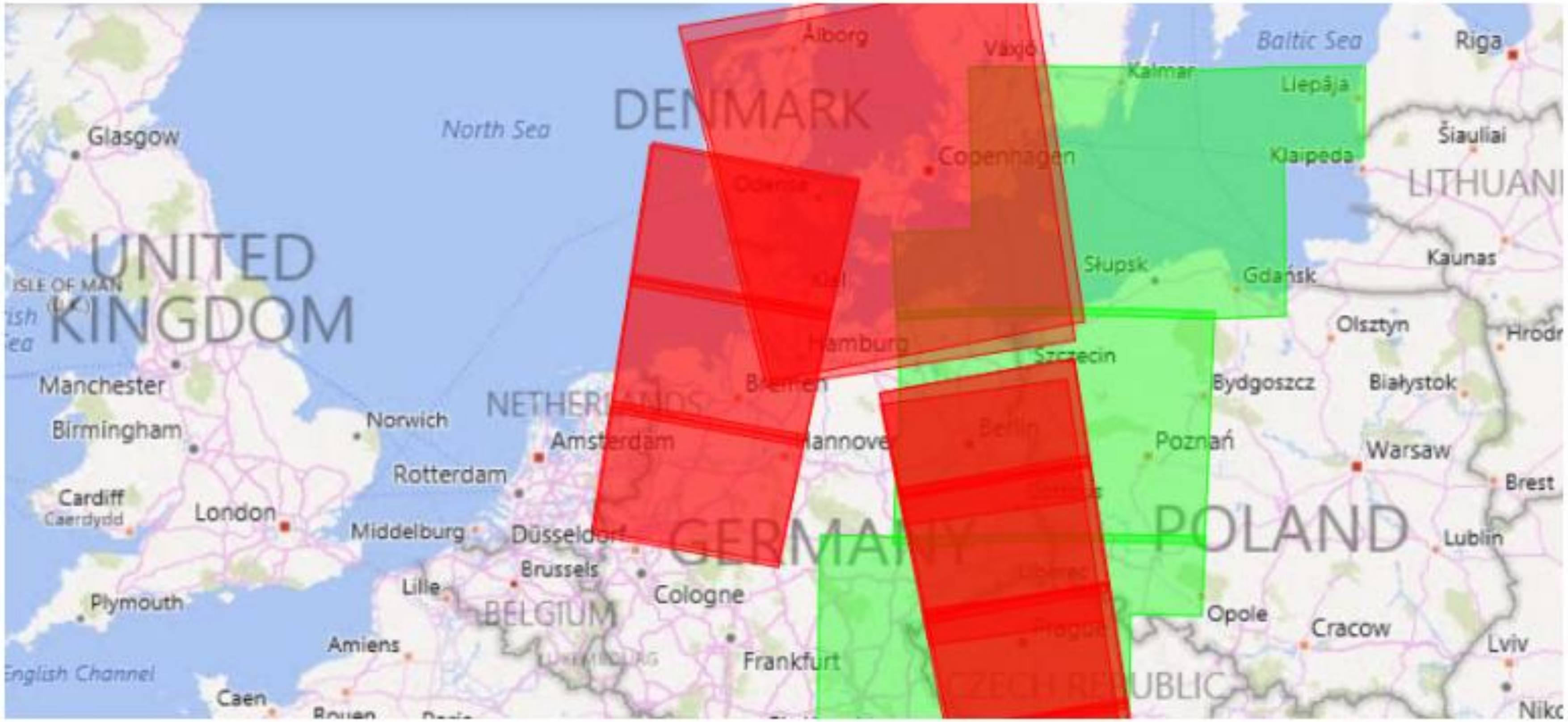
You need special software to process OSM raw data. It is usually command line and server software for Linux. Common tools to process OpenStreetMap raw data are:

Sentinel Scientific Data Hub.

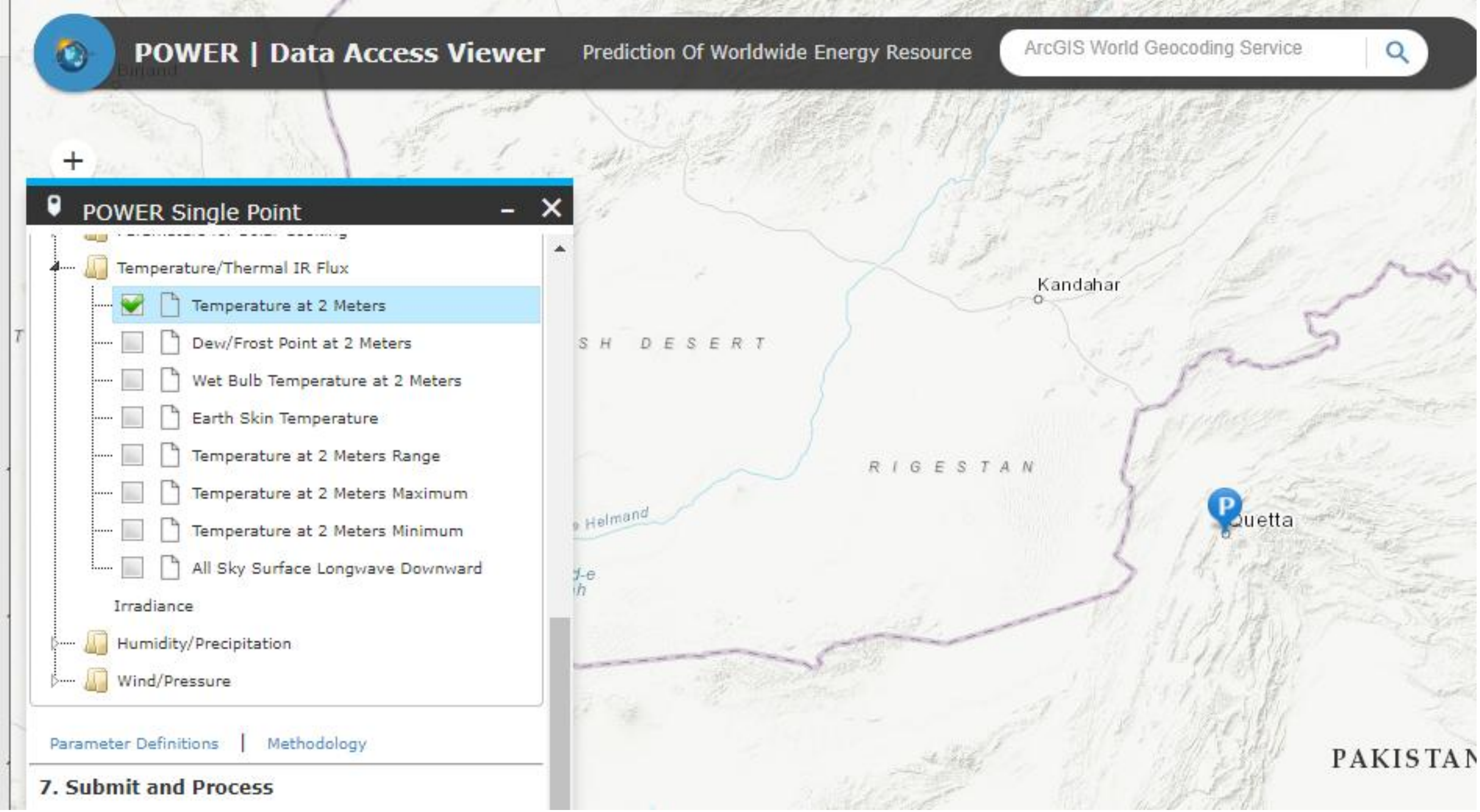
https://gisgeography.com/how-to-download-sentinel-satellite-data/

How to Download Free Sentinel Satellite Data

By: GISGeography • Last Updated: October 1, 2023

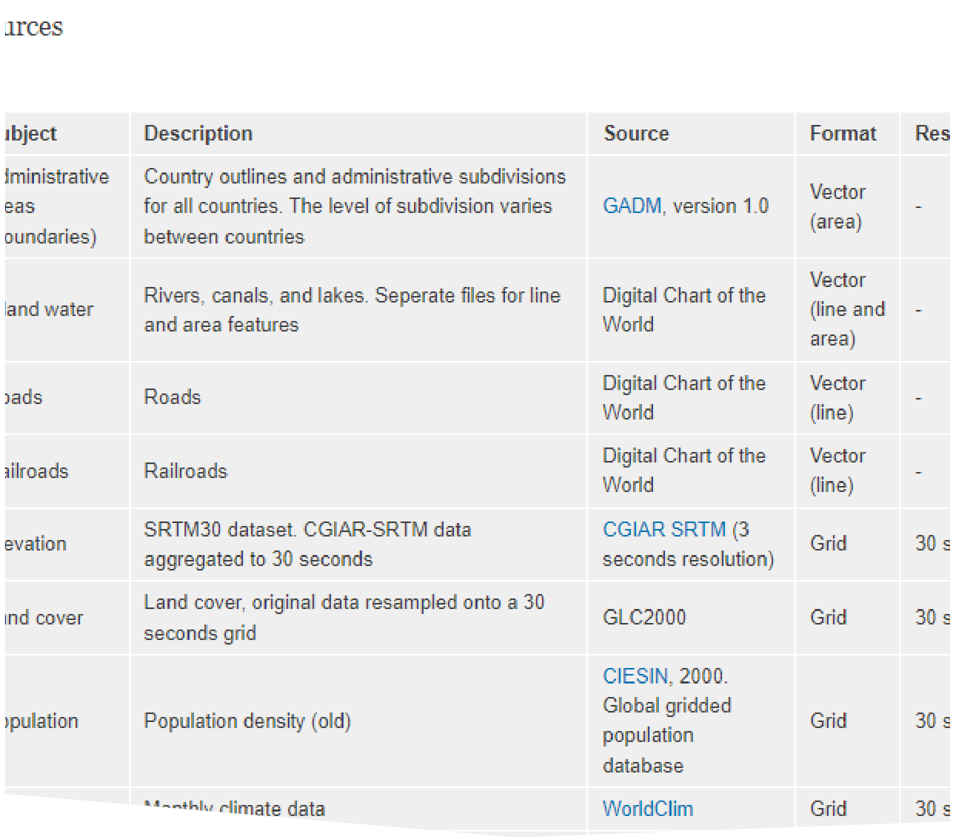


Download Sentinel-1 and Sentinel-2 Satellite Data



Weather Data

Link: https://power.larc.nasa.gov/data-access-viewer/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ıbject	Description	Source	Format	Res
dministrative eas oundaries)	Country outlines and administrative subdivisions for all countries. The level of subdivision varies between countries	GADM, version 1.0	Vector (area)	-
land water	Rivers, canals, and lakes. Seperate files for line and area features	Digital Chart of the World	Vector (line and area)	-
ɔads	Roads	Digital Chart of the World	Vector (line)	-
ailroads	Railroads	Digital Chart of the World	Vector (line)	-
evation	SRTM30 dataset. CGIAR-SRTM data aggregated to 30 seconds	CGIAR SRTM (3 seconds resolution)	Grid	30 s
ınd cover	Land cover, original data resampled onto a 30 seconds grid	GLC2000	Grid	30 s
ɔpulation	Population density (old)	CIESIN, 2000. Global gridded population database	Grid	30 s
	Monthly climate data	WorldClim	Grid	30 s

Diva-GIS:

- **Download Data by Country**

https://www.diva-gis.org/gdata

Diva GIS

Download Data by Country

https://diva-gis.org/data.html

DIVA-GIS

Data

« **Documentation** :: Contents :: **Climate data** »

Data

Select and download free geographic (GIS) data for any country in the world
 GIS data download

Country
Pakistan

Subject
Administrative areas

Format

Download **PAK_adm.zip**

Sources

Subject	Description	Source	Format	Resolution
Administrative areas (boundaries)	Country outlines and administrative subdivisions for all countries. The level of subdivisions that is available varies between countries	**GADM**	Vector (area)	-
Inland water	Rivers, canals, and lakes. Seperate files for line and area features	Digital Chart of the World	Vector (line and area)	-
Roads	Roads	Digital Chart of the World	Vector (line)	-
Railroads	Railroads	Digital Chart of the World	Vector (line)	-
Elevation	SRTM30 dataset. CGIAR-SRTM data aggregated to 30 seconds	**CGIAR SRTM** (3 seconds resolution)	Grid	30 seconds
Land cover	Land cover, original data resampled onto a 30 seconds grid	GLC2000	Grid	30 seconds
Population	Population density (old)	**CIESIN**, 2000. Global gridded population database	Grid	30 seconds
Climate	Monthly climate data	**WorldClim**	Grid	30 seconds
Gazetteer	A gazetteer is a list of place names and their coordinates. The files you can download here are for use in DIVA for automatic georeferencing (to assign coordinates to places). The files should be placed in the \\gazet directory. This is very old now. There area better services for this.	DBF	-	

Formats

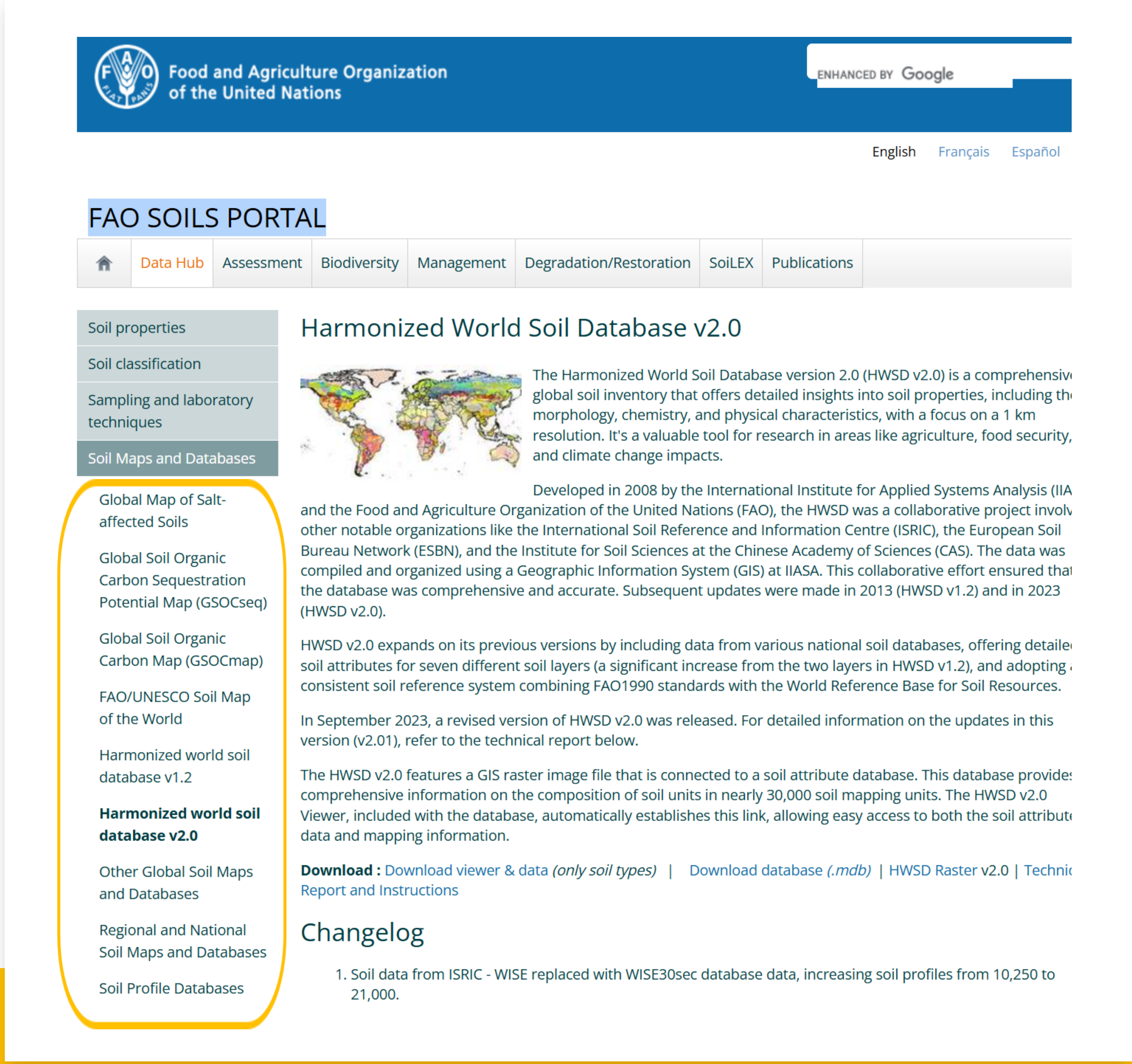
The files have been compressed and grouped in ZIP files. You can use programs such as 7-zip, PKZIP or StuffIt to decompress the files.

Vector data are stored as ESRI shapefiles Grid (raster) data are stored as DIVA gridfiles

FAO SOILS PORTAL

1 km resolution

https://www.fao.org/soils-portal/data-hub/soil-maps-and-databases/harmonized-world-soil-database-v20/en/


Food and Agriculture Organization of the United Nations

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FAO SOILS PORTAL

Data Hub | Assessment | Biodiversity | Management | Degradation/Restoration | SoiLEX | Publications

Soil properties

Soil classification

Sampling and laboratory techniques

Soil Maps and Databases

Global Map of Salt-affected Soils

Global Soil Organic Carbon Sequestration Potential Map (GSOCseq)

Global Soil Organic Carbon Map (GSOCmap)

FAO/UNESCO Soil Map of the World

Harmonized world soil database v1.2

Harmonized world soil database v2.0

Other Global Soil Maps and Databases

Regional and National Soil Maps and Databases

Soil Profile Databases

Harmonized World Soil Database v2.0

The Harmonized World Soil Database version 2.0 (HWSD v2.0) is a comprehensiv global soil inventory that offers detailed insights into soil properties, including th morphology, chemistry, and physical characteristics, with a focus on a 1 km resolution. It's a valuable tool for research in areas like agriculture, food security, and climate change impacts.

Developed in 2008 by the International Institute for Applied Systems Analysis (IIA and the Food and Agriculture Organization of the United Nations (FAO), the HWSD was a collaborative project involv other notable organizations like the International Soil Reference and Information Centre (ISRIC), the European Soil Bureau Network (ESBN), and the Institute for Soil Sciences at the Chinese Academy of Sciences (CAS). The data was compiled and organized using a Geographic Information System (GIS) at IIASA. This collaborative effort ensured tha the database was comprehensive and accurate. Subsequent updates were made in 2013 (HWSD v1.2) and in 2023 (HWSD v2.0).

HWSD v2.0 expands on its previous versions by including data from various national soil databases, offering detaile soil attributes for seven different soil layers (a significant increase from the two layers in HWSD v1.2), and adopting consistent soil reference system combining FAO1990 standards with the World Reference Base for Soil Resources.

In September 2023, a revised version of HWSD v2.0 was released. For detailed information on the updates in this version (v2.01), refer to the technical report below.

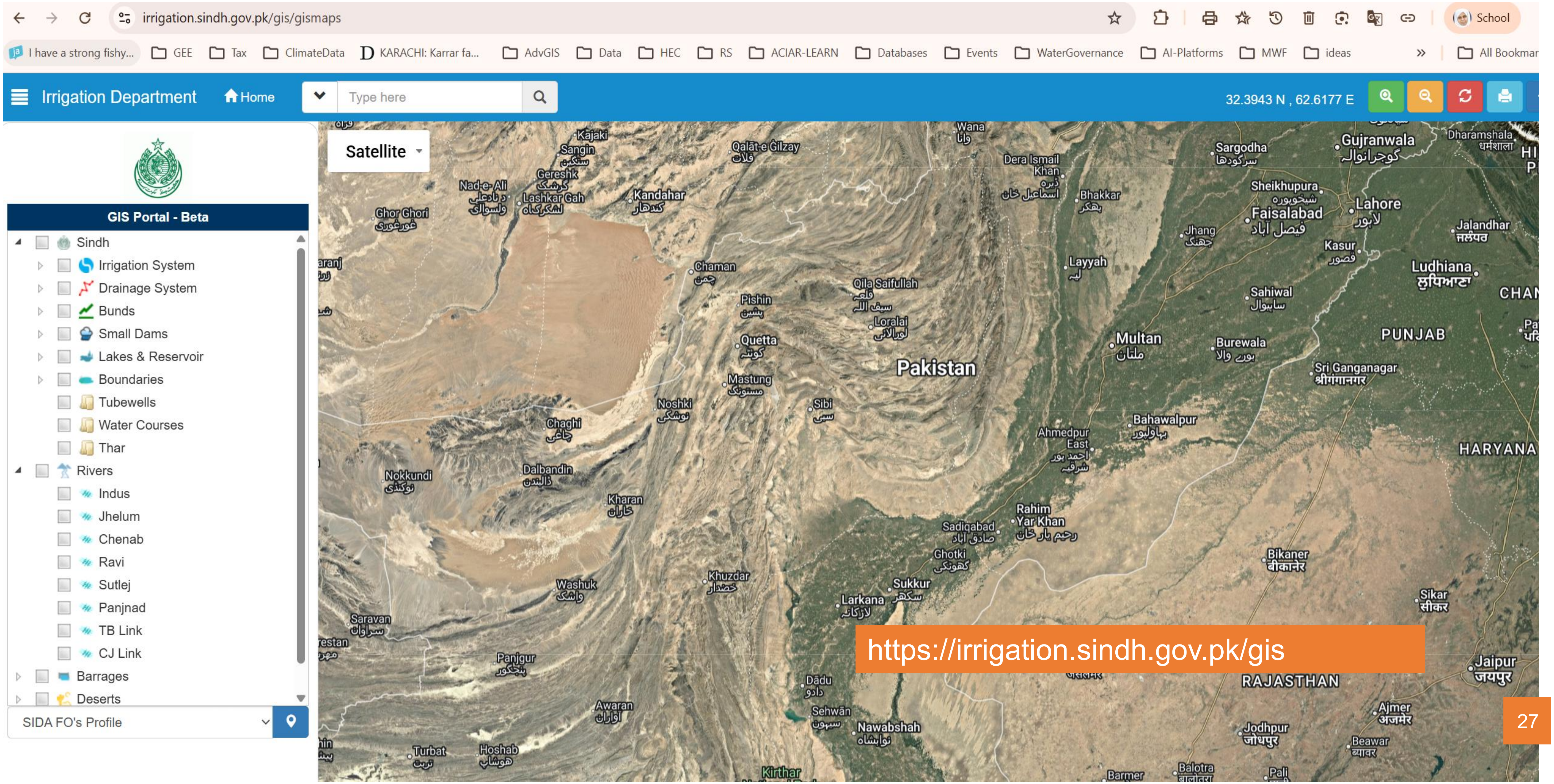
The HWSD v2.0 features a GIS raster image file that is connected to a soil attribute database. This database provide comprehensive information on the composition of soil units in nearly 30,000 soil mapping units. The HWSD v2.0 Viewer, included with the database, automatically establishes this link, allowing easy access to both the soil attribut data and mapping information.

Download : Download viewer & data *(only soil types)* | Download database *(.mdb)* | HWSD Raster v2.0 | Technic Report and Instructions

Changelog

1. Soil data from ISRIC - WISE replaced with WISE30sec database data, increasing soil profiles from 10,250 to 21,000.

Sindh Irrigation GIS Portal



IRRIGATION DEPARTMENT

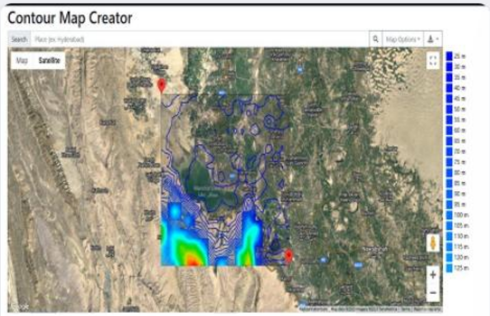
Home | About | Team | Attached Bodies | Sindh Water Policy | Water | E-Library | Events | Contact

CLOUD APPS GALLERY

Experimental

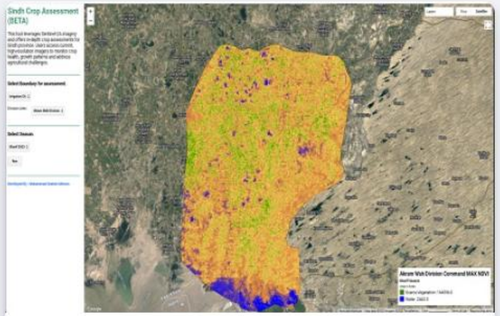
Irrigation GIS Portal

The GIS Portal offers comprehensive geospatial data, facilitating informed decision-making for efficient water resource management.



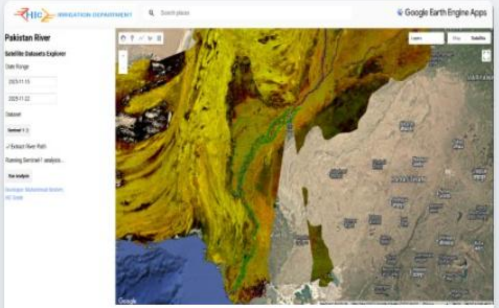
Contour Mapping

This app allows efficient visualization and analysis of land elevations, aiding in planning and water evacuation.



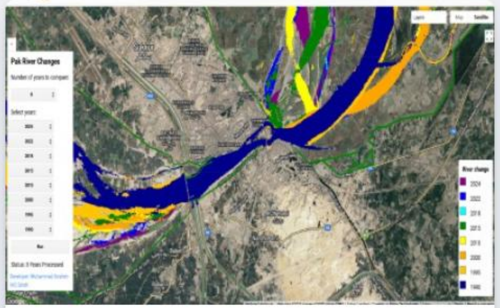
Sindh Crop Assessment

This tool leverages Sentinel-2A imagery and offers in-depth crop assessments for Sindh province.



Pak River Path

This tool leverages Sentinel-2A and Landsat imagery and offers in-depth Pak rivers .

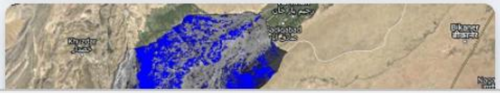


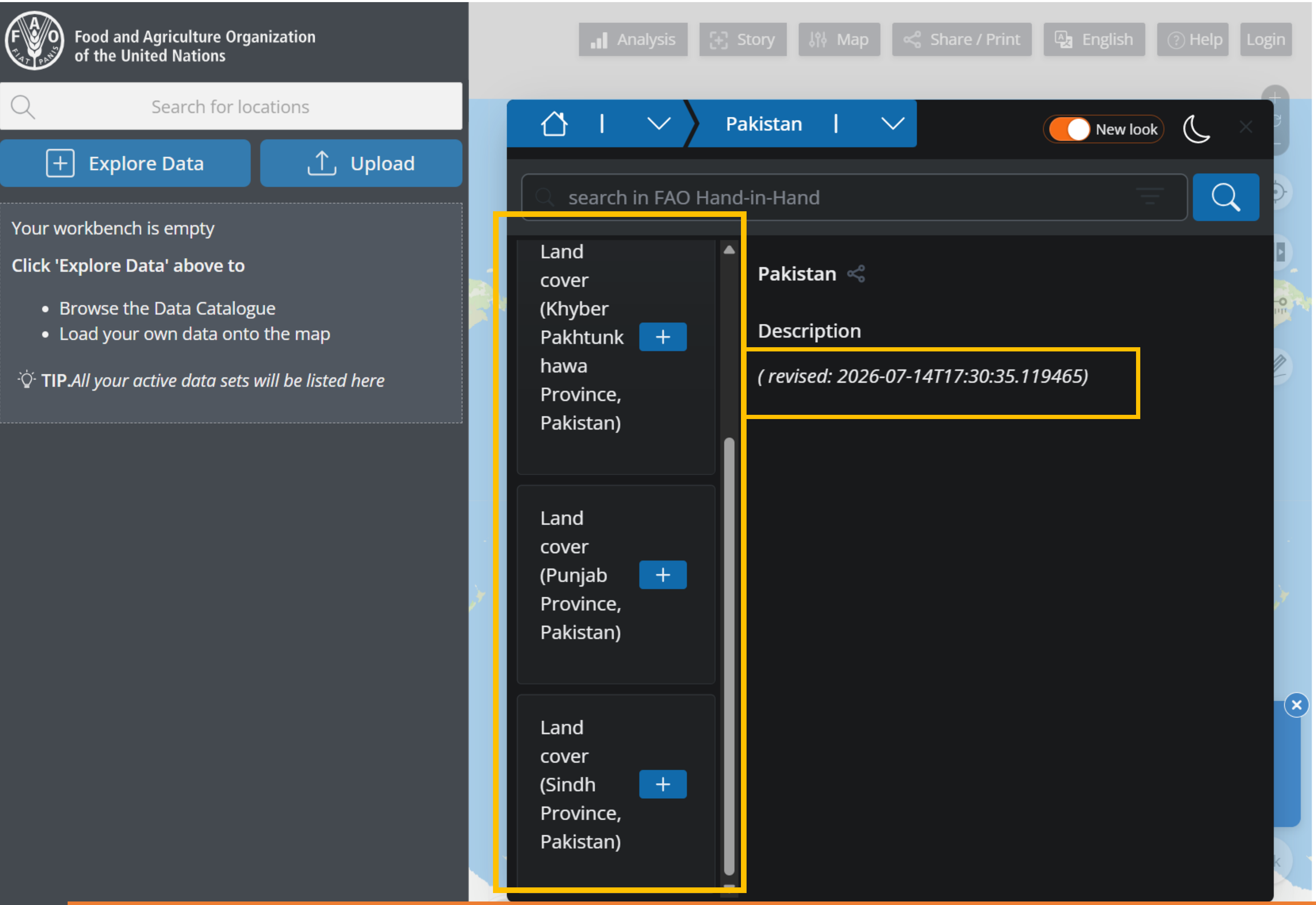
Pak River Path

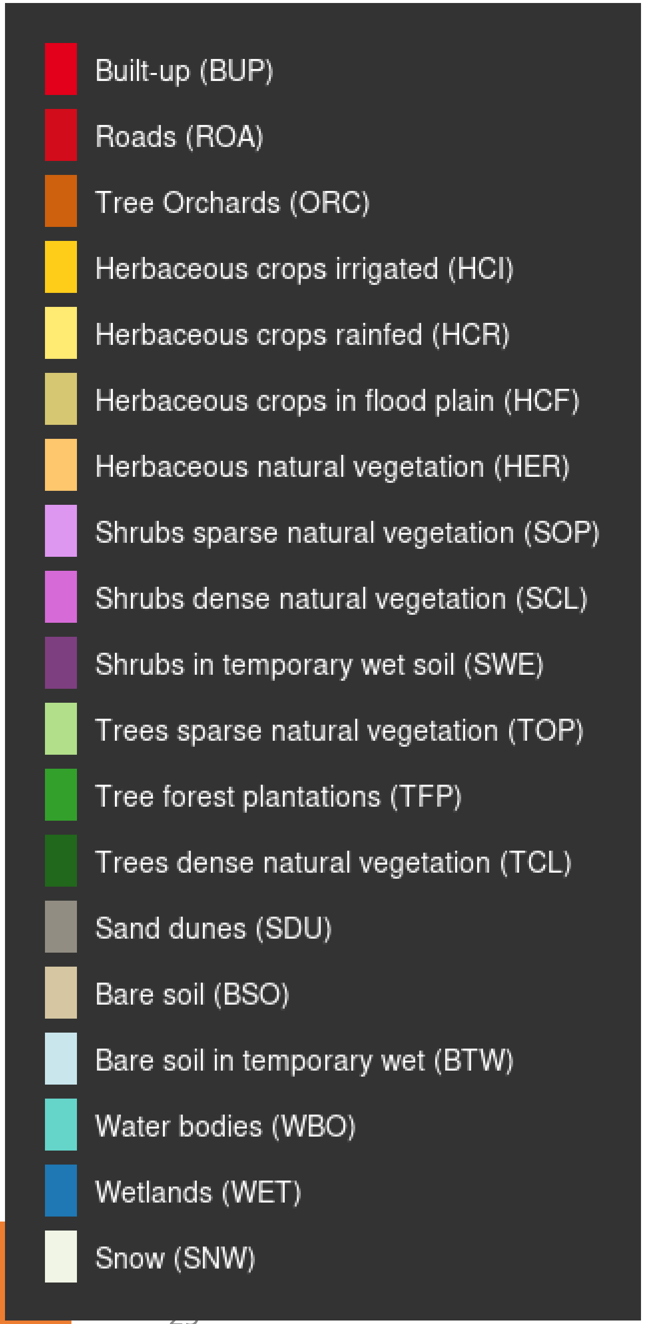
This tool leverages Landsat imagery and offers changes in rivers path .

ACWA Portal Beta

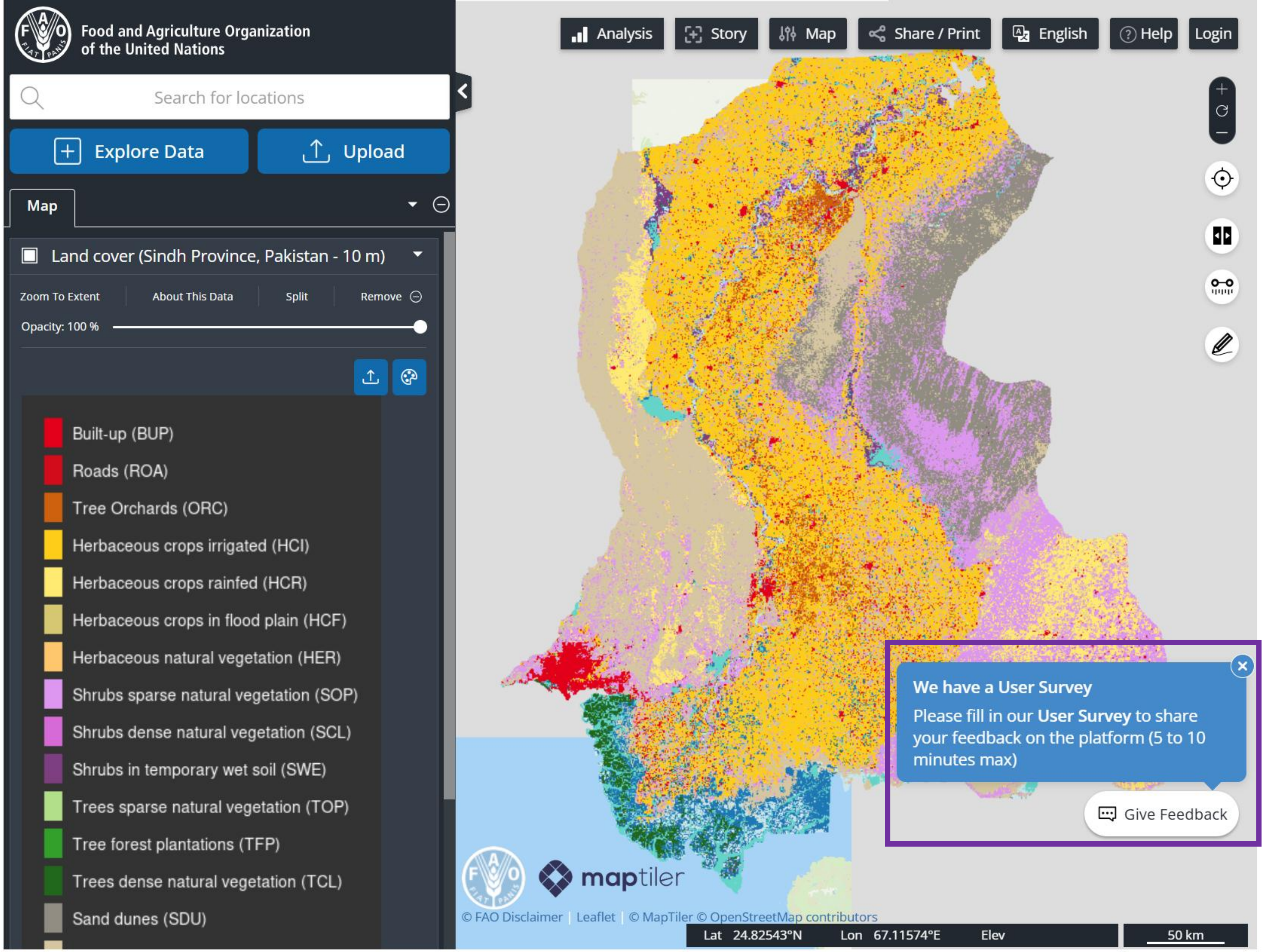
GCF-FAO Project ACWA Portal combines field and remtoe sensing data.





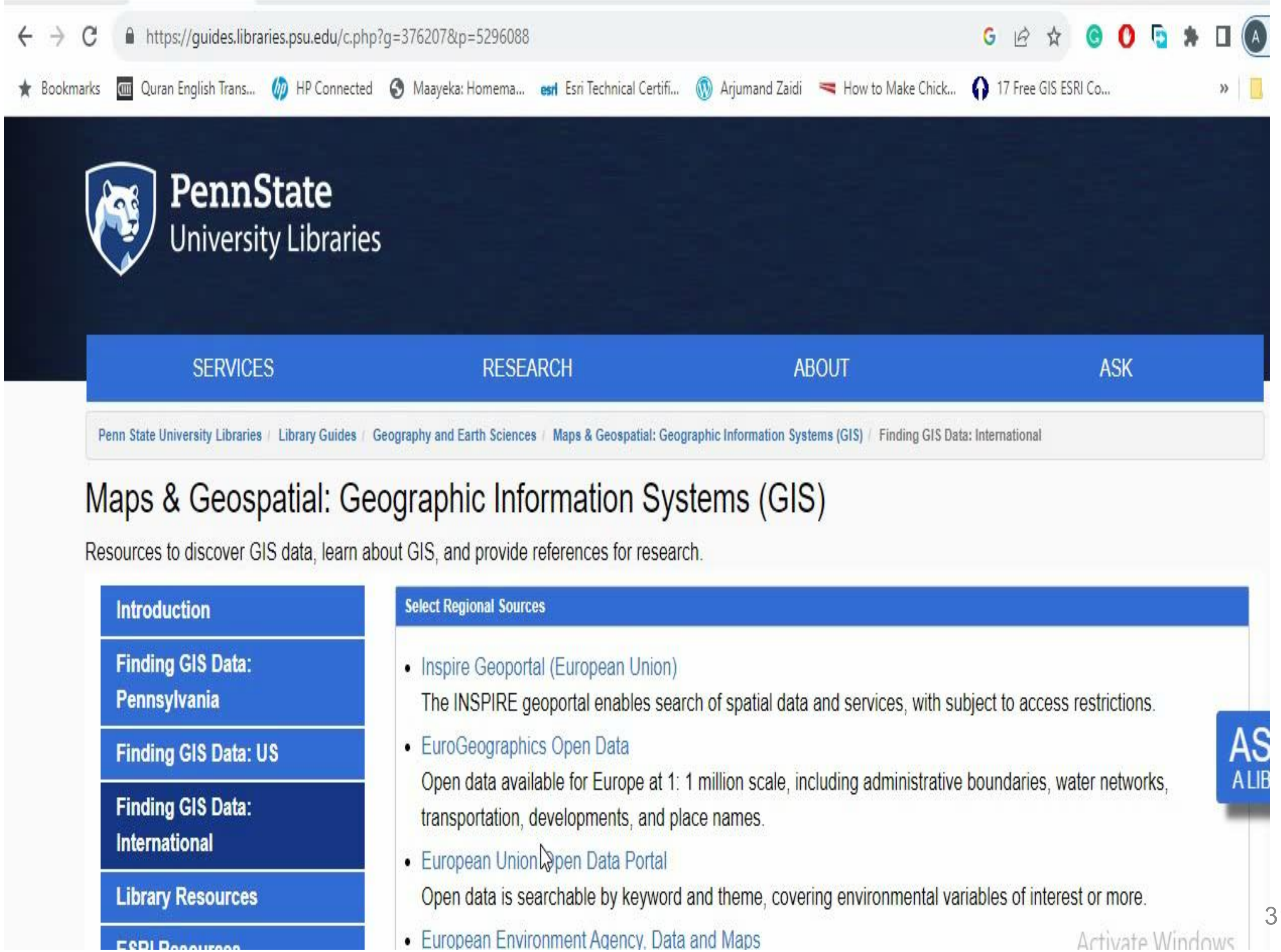


Land cover (Provinces of Pakistan - 10m)

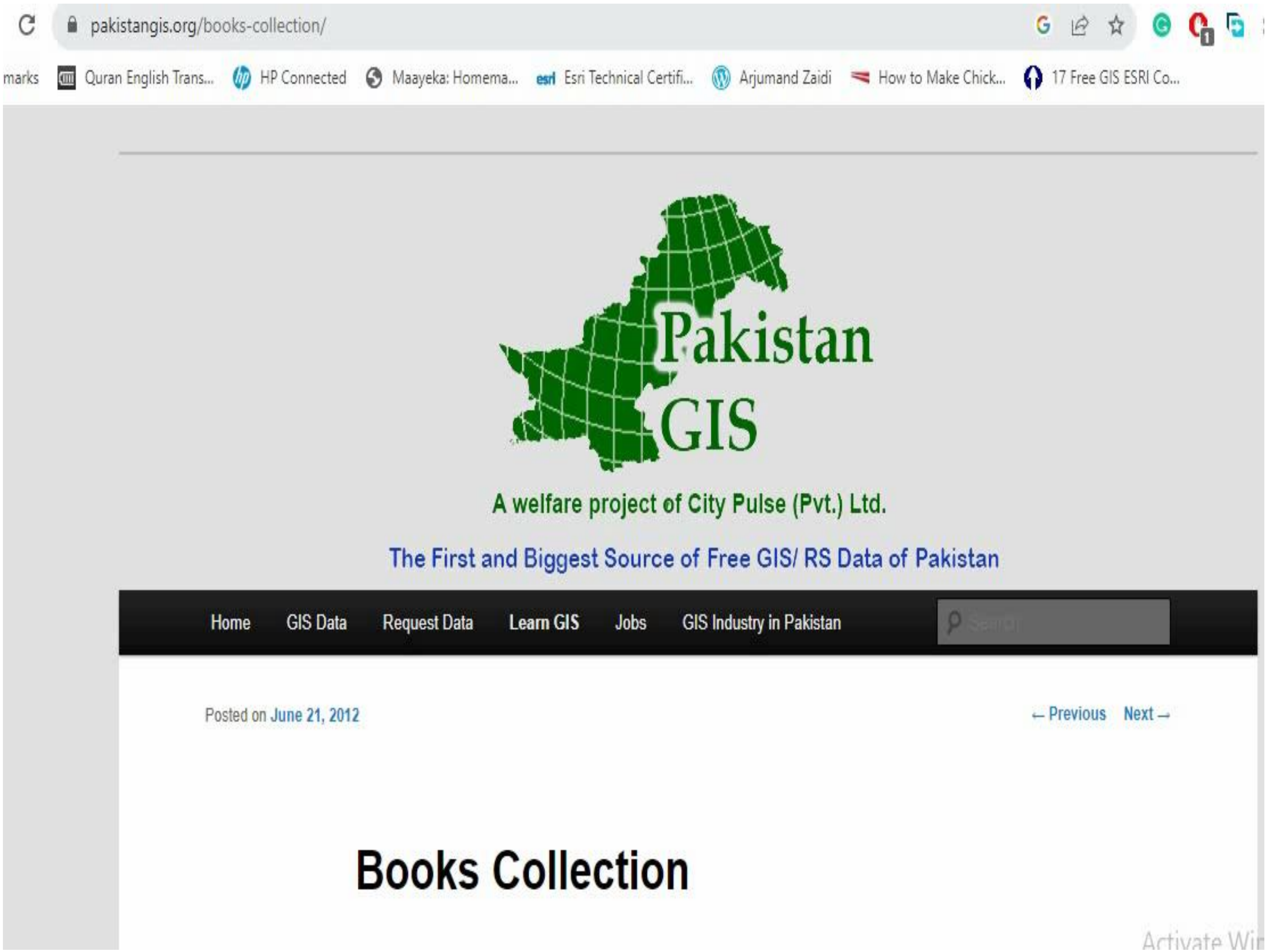


Explore this site and fill out the survey form

Other sources



Data For Pakistan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HP Connected
Maayeka: Homema...
Esri Technical Certifi...
Arjumand Zaidi
How to Make Chick...
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GIS
A welfare project of City Pulse (Pvt.) Ltd.
The First and Biggest Source of Free GIS/ RS Data of Pakistan
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